FEATURE 05 OF 14 | PRIORITY P0

Message Highlighting

Priority	P0
Judging criteria served	Innovation (20) | UI/UX (15)
Estimated effort	Medium (~2-4 hrs) (rough estimate)
Depends on	Rule-Based Detection

Purpose

Re-displays the user's own message with the exact suspicious phrases and links marked. Seeing the danger inside their own text is far more convincing than a score alone, and it is the clearest visual proof of the explainable approach.

What it must do

- Show the original message with every matched phrase and URL highlighted.
- Different categories use different highlight styles (colour plus underline style), with a small legend.
- Tapping/hovering a highlight shows the rule name; clicking it scrolls to the matching card in the explanation panel (feature 6).
- Overlapping matches (e.g. a phrase that triggers two rules) are merged into one visible span carrying both labels.
- Output must be XSS-safe: build the highlighted text from text nodes / escaped strings only.

How it should work

- **Algorithm:** 1) collect all matches as {start, end, ruleIds}; 2) sort by start; 3) merge overlapping or touching ranges, combining ruleIds; 4) walk the original string, emitting plain text for gaps and a <mark> element for each merged range.
- Offsets always refer to the original pasted string (see feature 1). Test with emoji and Hindi characters to confirm offsets are not shifted.
- Use semantic <mark> elements with an aria-label such as "Fake urgency: act immediately" so screen-reader users get the same information.
- Never use innerHTML with user text. Create text nodes via the DOM, or escape &, <, >, quotes before building any HTML string.

UI / UX notes

- Use underline style differences (solid, dashed, wavy) in addition to colour so categories are distinguishable without colour.
- Keep line-height generous (about 1.6) so highlighted text stays readable on mobile.
- Legend chips above the message: Urgency, Threat, Link, Credential request.
- Long messages scroll inside a bounded box; the score and level stay visible.

Edge cases and risks

- Two rules match the same words - show a single mark with two labels, not nested broken markup.

- A URL that is also part of a threat sentence - the URL highlight wins in the overlap region only if you define priority; recommended: merge into one span.
- Message with line breaks: preserve them (use white-space: pre-wrap).
- No matches: show the message plainly with 'No warning phrases found'.

Acceptance tests

Try this	Expected result
Example SBI message	4 highlighted regions, legend shows 4 categories
Message with text Hi	Shows literal Hi, not bold text
Overlapping match 'act immediately or account blocked'	One merged span, two labels, no broken markup
Message with emoji before a match	Highlight lands on the correct words
Click a highlight	Page scrolls to the matching explanation card

Demo tip for the judges

> Paste a message and let the judges watch the red phrases light up - this is the most visually memorable moment of the prototype.